



Pragmatic Environmentalist of New York

Balancing the risks and benefits of environmental initiatives

PM2.5 Health Impacts in New York City

In the last several days I have been drafting a review of the [PEAK Coalition](#) report entitled: “[Dirty Energy, Big Money](#)” and today I was working on the air quality health impacts section. I also noticed today that the usual suspects are claiming [links between air pollution and Covid-19](#) susceptibility. In this post I will explain how I could be convinced that the reports underlying presumption that inhalable particulates have dire health impacts is correct.

I am a retired electric utility meteorologist with nearly 40 years-experience analyzing the effects of air quality and meteorology on electric operations. I have been reviewed health impact claims throughout my career. This background served me well preparing this post. The opinions expressed in this post do not reflect the position of any of my previous employers or any other company I have been associated with, these comments are mine alone.

Background

Health impacts associated with inhalable [particulates](#), also known as PM2.5 because it refers to airborne particles with a diameter of 2.5 micrometers or less, turn out to be the primary rationale for all the recent EPA air quality emission reductions cost-benefit analyses. For example, EPA’s air toxics emission limits were cost effective

not because of direct impacts of mercury and other heavy metals but because the control systems for those pollutants would have decreased PM_{2.5} concentrations and led to alleged health improvements.

Steve Milloy's [Scare Pollution: Why and How to Fix the EPA](#) explains the problems with those health impact claims. Milloy points out that no one has proven a biological explanation why the inhaled particles will cause fatal inflammation. The alleged relationship is based on epidemiological statistical evaluation of air quality and health impact data. The basic problem is that there are many confounding factors known to cause the observed health impacts and trying to tease air quality impacts out of the mix is difficult to prove.

It gets worse. The studies that are the basis for the alleged air quality health impacts were at relatively high ambient concentrations. Make no mistake that air pollution can be a very bad thing but the levels of pollution in the United States that clearly caused health impacts occurred many years ago and included a mix of pollutants not found anywhere in this country today. It gets worse because the dose–health impact relationship is being extrapolated using the [linear no–threshold model](#) which has been used to estimate the dose response for radiation health impacts. The concept is that there is no threshold below which there is no effect. However, in my opinion and [others](#), extrapolating measurements and responses at high levels down to levels near the level of detection is an unwarranted presumption. Nonetheless, advocates for ever lower air quality improvements routinely claim health impacts behave the same way.

Public Health Impacts

The primary public health reference in the PEAK Coalition report I am reviewing was the New York City Department of Health and Mental Hygiene's (DOHMH) [Air Pollution and the Health of New Yorkers report](#). The PEAK coalition description of air quality public health impacts quotes the conclusion from the DOHMOH report: "Each year, PM_{2.5} pollution in [New York City] causes more than 3,000 deaths, 2,000 hospital admissions for lung and heart conditions, and approximately 6,000 emergency department visits for asthma in children and adults." These conclusions

are for average air pollution levels in New York City as a whole over the period 2005–2007.

The DOHMOH report specified four scenarios for comparisons ([DOHMOH Figure 4](#)) and calculated health events that it attributed to citywide PM_{2.5} ([DOHMOH Table 5](#)). Based on their results the report notes that:

Even a feasible, modest reduction (10%) in PM_{2.5} concentrations could prevent more than 300 premature deaths, 200 hospital admissions and 600 emergency department visits. Achieving the PlaNYC goal of “cleanest air of any big city” would result in even more substantial public health benefits.

It is important to note how air quality has improved since the time of this analysis. The [NYS DEC air quality monitoring system](#) has operated a PM_{2.5} monitor at the Botanical Garden in New York city since 1999 so I compared the data from that site for the same period as this analysis relative to the most recent data available ([Data from Figure 4. Baseline annual average PM_{2.5} levels in New York City](#)). The Botanical Garden site had an annual average PM_{2.5} level of 13 µg/m³ for the same period as the report’s 13.9 µg/m³ “current conditions” city-wide average (my estimate based on their graph). The important thing to note is that the latest available average (2016–2018) for a comparable three-year average at the Botanical Garden is 8.1 µg/m³ which represents a 38% decrease. That is substantially lower than the PlaNYC goal of “cleanest air of any big city” scenario at an estimated city-wide average of 10.9 µg/m³.

Note that in [DOHMOH Table 5](#) the annual health events for the 10% reduction and “cleanest” city scenarios are shown as changes not as the total number of events listed for the current levels scenario. My modified table ([Modified Table 5. Annual health events attributable to citywide PM_{2.5} level](#)) converts those estimates to totals so that the numbers are directly comparable. I excluded the confidence interval information because I don’t know how to convert them in this instance.

I confirmed that the DOHMOH analysis used a linear no-threshold health impact analysis and used their relationship to estimate the effect of the observed air quality reduction. I tested the linear hypothesis by scaling the “current level” scenario

number of events to the proportion of the PM 2.5 concentrations (the last row in the table) for the “current level” and the other two scenarios. My estimated health impacts were all within 1% which proves that the DOHMOH analysis relied on a linear no-threshold approach. As a result, that means that I could estimate the health impact improvements due to the observed reductions in PM2.5 as shown in the last three columns in the modified table. I estimate that the observed reduction in PM2.5 concentrations prevented nearly 1,300 premature deaths, 800 hospital admissions and 2400 emergency department visits.

Conclusion

In order to convince me that the PM2.5 health impacts claimed by MOHDOH and many others are correct I need to see confirmation with observed data. The DOHMOH report claims that in 2005-2007 that PM2.5 concentrations led to, for example, 3,200 premature mortality events. I have no idea how that number compares to observed values for this parameter or the others included. I estimate that for the observed reductions in measured PM2.5 the number of premature mortality events would be reduced 1,296 events down to 1,904 events.

The first question for the health experts is whether the change from 2005-2007 to 2016-2018 of 1,296 events could be observed against natural variations or is that number within the normally expected variation. If not then my hope for verification is not possible but more importantly it also means that the gloom and doom stories of significant health impacts are base on nothing more than insignificant statistical noise that is not really observable. If those data are greater than expected natural variation, then it would be possible to document improvements in these alleged health impacts due to the 38% decrease in PM2.5. If that is the case, then I stand corrected.

Here is the thing though. The percentage of [people with asthma in the United States from 2001 to 2018](#) is not showing a decrease at the same time [ambient levels of all air pollutants are going down](#) substantially. While correlation does not necessarily mean causation, no correlation with a purported cause indicates a bet on a dead horse. Therefore, I am not holding my breath that the data will show the purported benefits.



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I am a meteorologist (BS and MS degrees), was certified as a consulting meteorologist and have worked in the air quality industry for over 40 years. I author two blogs. Environmental staff in any industry have to be pragmatic balancing risks and benefits and (<https://pragmaticenvironmentalistofnewyork.blog/>) reflects that outlook. The second blog addresses the New York State Reforming the Energy Vision initiative (<https://reformingtheenergyvisioninconvenienttruths.wordpress.com>). Any of my comments on the web or posts on my blogs are my opinion only. In no way do they reflect the position of any of my past employers or any company I was associated with.

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2 thoughts on “PM2.5 Health Impacts in New York City”



slow to follow

May 20, 2021 at 10:04 am

You might be interested in this web event next week May 25, 2021:

<https://www.healtheffects.org/annual-conference>

Webinar 8: What We Have Learned about Effects on Health at Low Levels of Exposure: Evidence from the United States, Canada, and Europe

May 25, 2021 – 10:00am

Chairs: Amy Herring, Duke University, HEI Research Committee, and Sverre Vedal, University of Washington

In 2015, HEI funded three studies to investigate the health effects of long-term exposures to low levels of air pollution in very large populations across the United

States, Canada, and Europe. This session will present results from these recently completed studies, discuss their strengths and weaknesses, and discuss potential implications for future risk assessment and regulation. Highlights include thorough examinations of concentration–response functions, results for multiple health endpoints (e.g., cause-specific mortality, respiratory events), as well as multiple pollutants (e.g., PM_{2.5}, O₃, NO₂), and causal modeling approaches.

Register now

10:00 AM Introduction

10:10 AM Effects of Confounder Control and Causal Modelling

Danielle Braun

Harvard University

10:25 AM Comparison of Concentration–Response Functions

Tanya Christidis

Statistics Canada

10:40 AM Results from Multi-Pollutant Models

Jie Chen

Utrecht University

10:55 AM Commentary and Critique

Sverre Vedal

University of Washington, Seattle

11:10 AM Questions and Discussion

https://zoom.us/webinar/register/WN_yOTkRsSPT1Ks4CosHWAmhw



rogercaiazza 

May 20, 2021 at 1:20 pm

Thank you. I am interested.



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